

Therapet: Designing an Interactive XR Therapeutic Pet

Jordan Glendinning
University of Abertay

Haocheng Yang
University of Abertay

Andrew Hogue*
Ontario Tech University

Patrick C. K. Hung
Ontario Tech University

Ruth Falconer†
University of Abertay

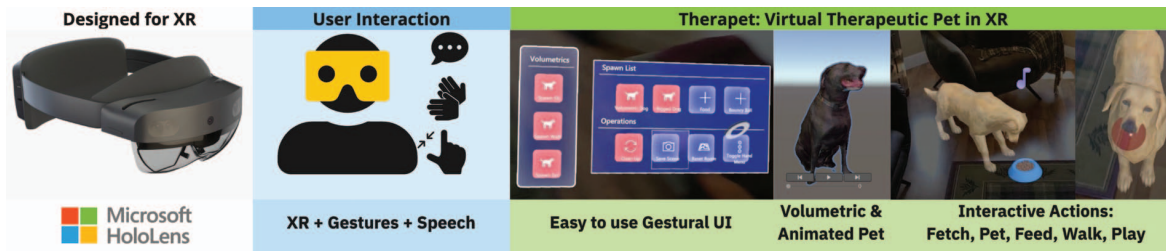


Figure 1: Overview of our Therapeutic Pet Prototype System.

ABSTRACT

As Extended Reality (XR) technologies continue to mature, they offer ever more diverse, complex, and impactful experiences. This paper presents “Therapet”, an augmented reality (AR) application designed for AR glasses such as the HoloLens 2. Therapet presents an interactive therapeutic pet avatar—using both Volumetric Video (VV) recordings and traditionally animated rigged models—with which users can interact with by feeding, playing fetch, and petting. We conducted a pilot study to investigate usability and user preferences, revealing positive user experiences and insights into the use of (and perceptions towards) Volumetric Video based avatars for interactive AR experiences. Furthermore, we analyze the feasibility of using VV in production (where memory usage and performance are critical factors) and discuss lessons learned to create a successful AR therapeutic pet application.

Index Terms: Human-centered computing—Visualization—Visualization design and evaluation methods Applied Computing—Law, social and behavioral sciences—Psychology

1 INTRODUCTION AND RELATED WORK

As Augmented Reality (AR) glasses approach ever closer towards affordability in the consumer space, applications for augmenting the world are becoming ever more prevalent. One potential use-case for AR is the reduction of anxiety for students during exam times at post-secondary institutions. This has the potential to reduce anxiety levels (for those who are not afraid of animals) during stressful periods. Ward et al [4] recently assessed the effectiveness of therapy dog programs on university campuses and found strong immediate benefits from interacting with therapy dogs, including reduced stress and increased happiness and energy levels.

Virtual Pets for Therapy. The development of a virtual pet application designed for use in similar scenarios allows for the examination of whether virtual representations of animals have similar effects on reducing anxiety. The use of a virtual pet in this way could have significant impact on student-populations especially at

institutions with strict policies regarding animals on campus. Moreover, one could imagine using a virtual pet application to supplement single, drop-in therapy sessions potentially providing a longer-term impact. It has been shown that individuals respond to virtual pets psychologically and physiologically. For example, Freeling [2] established that a strong sense of embodiment is closely linked to intense emotional experiences when interacting with a virtual cat. In this paper, we describe the design of a therapy-pet augmented reality application where the user can select between two different types of visual representations of the ‘Therapet’. The prototype was developed for the HoloLens2 and features gameplay mechanics for interacting with the virtual pet such as feeding, fetching and petting. We present results from a pilot exploring user perception of these virtual pets and the system usability to inform future development of this project.

Volumetric Video. Since the initial concept of ‘Virtualized Reality’ [3] was introduced in 1995, volumetric video has evolved into a new medium promising realistic 4D video performances. Traditionally, VV is captured using a dedicated studio setup such as those from 4DViews (Holosys), 8i, or Metastage all of which require a significant investment of a large dedicated space and potentially hundreds of cameras to capture the performance volume accurately. Alternatively, DepthKit Studio (Scatter), Evercoast, Soar, and FormaVision are examples of prosumer turn-key volumetric solutions that can record or stream 4D performances and integrate into modern game engines relatively easily. It is worth noting that while these solutions are effective at capturing high quality VV assets, their high costs and opaque algorithms present entry barriers that can discourage VV creators to experiment. The Therapet application uses a commercial multi-camera Red Green Blue-Depth (RGB-D) capture solution capable of recording non-human performances (Soar).

Therapet Design. The primary goal was to design an AR application to present the user with opportunities for interaction—and to compare user reactions to—two visually distinct types of virtual pets (Photorealistic (PR) versus Non-photorealistic (NPR)). During the design phase and requirements analysis, we informally engaged a number of pet owners to determine the various possible ways in which a user could effectively interact with a virtual pet and how best to provide appropriate user feedback. This resulted in three distinct activities that pet owners would commonly participate in with their dog. **1: Feeding:** Users are able to ‘feed’ their Therapets by spawning a “dog bowl”. **2: Petting.** Users are able to illicit a response from their Therapet via a ‘petting’ gesture. **3: Fetching.**

*e-mail: andrew.hogue@ontariotechu.ca

†e-mail: r.falconer@abertay.ac.uk

User	Score	Grade	Adjective	SUS LEGEND		
P1	72.5	B	Good	>80.3	A	Excellent
P2	80	B	Good	68-80.3	B	Good
P3	60	D	Poor	68	C	Okay
P4	75	B	Good	51-68	D	Poor
P5	80	B	Good	<51	F	Awful

Table 1: SUS Total Scores & Legend

Users are able to ‘play’ fetch with the virtual dog. In this mode the virtual pet will use the NavMesh to plan the appropriate path through the scanned world to pick up the ball and bring it back to the user. **Visualized Emotions.** A small set of emoticon visual effects (emojis) enhance user feedback during user-pet interactions.

Pilot Study. A small pilot study was conducted to assess the virtual pet assets and highlight potential improvements. The study consisted of 5 volunteers who all fell in the age range of 16 to 28. For the purposes of the study, the Therapet application allowed users to spawn rigged-mesh (NPR) and VV (PR) versions of the pet avatars. Participants were given a short training session to become familiar with the interface and learn how to interact with the application using gestures and voice commands. Participants were then prompted to observe a Therapet instance located at the centre of the playspace and asked to identify the animals’ species. The type of Therapet (rigged-mesh or VV) was randomly selected (counter-balanced) before the start of each play session to remove any ordering effects. Participants were instructed to interact with the Therapet – using all available interactions methods – for a period of time. Afterwards each participant completed a “System Usability Scale” (SUS) [1] questionnaire and were debriefed on their experience with the application. **Analysis.** The results of the SUS questionnaire show that participants reacted positively overall to the application. Most participants indicated that they would like to use the application more frequently, potentially due to a combination of their overall confidence with the application and their belief that the system was easy-to-use. Despite users finding the system simple to use, some participants also indicated that they would require the support of a technical person to use the application. Over the course of the study, 2 out of the 5 participants made explicit mention of their preference for the photo-realistic VV asset over the rigged-mesh, while another 2 participants preferred the traditional rigged-mesh over the VV assets. Participants who preferred the rigged-mesh appearance of the Therapet often cited its vivid appearance and the movement of its tail, while the appearance of the VV asset was described as “distorted” and “horrifying” due to the “jittering” appearance of its vertices as well as not being able to clearly see the Therapets’ eyes – a byproduct of the recording and optimisation processes. Those who preferred the VV assets claimed its appearance was much closer to resembling a real dog and made explicit reference to the pets’ breathing during its idle animation, with one participant describing the Therapet as cuter compared to its rigged-mesh counterpart. The SUS Total Scores were computed as in [1]. 4 out of the 5 participants rated the interface overall in the *Good* range while one rated it as *Poor*. The major differences in this participants’ responses are that they felt the system functions needed to be integrated better and that the system is too complex.

Discussion. One of the primary goals of the initial pilot study was to gain insight into how users would react to VVs compared to traditional rigged meshes. Our study showed that the while VVs offer much more accurate depictions of pets through subtle movements, the current implementation was limited by the lack of visual clarity and inconsistent quality due to the necessary post-processing for the HoloLens 2, which can cause them to appear uncanny to some users. Participants’ feedback for VV assets, mainly focused around inconsistent vertex placement and low texture resolution. While the low resolution textures of a VV can be solved by modifying the import settings during the asset preparation process, the quality of textures are limited only by the available memory of the target platform. A

potential solution to this issue could be to stream VV textures from memory or to employ a more complex memory-management system to handle the loading and unloading of textures at runtime. Creating more stable VV animations is a significantly harder task to achieve due to the “jittering” effect being a byproduct of the initial recording process. While vertex displacement can be somewhat mitigated through the series of post-processing steps outlined above, a more rigorous vertex-stabilization could help reduce the distortion effect present on VV assets.

Pilot Study and System Usability. While user responses from the pilot study show a generally positive reaction to the Therapet application, data suggests that a more refined user interface is required to improve the overall user experience. Responses to the study show that while users enjoyed using the Therapet application, some users found the User Interface (UI) was needlessly complex and limited the overall user experience. The Therapet UI currently relies on interaction through specific “finger-pointing” gestures which, during the pilot study, would often struggle to accurately identify participants’ fingers. Half of all participants described the system as relatively cumbersome to use, those same participants also show to be lacking confidence in their ability to use the Therapet system. By comparing this data to observations made during the study, it could be said that the system used to interact with virtual objects greatly impacts a users ability to effectively use UI elements and virtual objects and as a result, struggle to use the application as a whole. This could be attributed to the gesture-based interaction of the HoloLens 2. Although we expected users would struggle due to the lack of haptic feedback from the in-air hand-tracking input, we observed that the majority of participants struggled when picking-up items to interact with.

2 CONCLUSIONS AND FUTURE WORK

Therapet, a prototype AR application with the long-term aim to mitigate anxiety through meaningful interactions with a virtual pet dog was presented. The pilot gave a preliminary understanding of user engagement, preferences, and reactions to the pet assets in terms of photorealism and animation. It was identified that the post-processing of the volumetric assets (required to meet technical limitations of the headset) compromised the animation quality, causing some users to disengage. This highlights a critical limitation that necessitates further development of optimization and animation techniques for volumetric video sequences. The next phase of this research will focus on improving the gesture system as well as the optimization and animation methods for VV assets. Longer term we will employ a randomized control trial to measure the impact of interactivity and realism on users’ anxiety levels and compare these effects with traditional physical pet interactions.

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